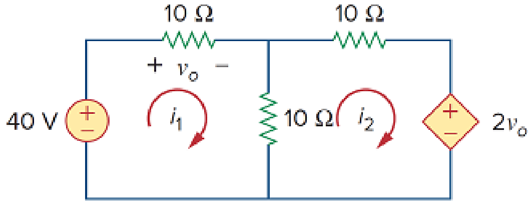
Chapter 3, Problem 46P

Problem

Calculate the mesh currents *i*1 and *i*2 in Fig. 3.92.

Figure 3.92:



Step-by-step solution

Step 1 of 4

Refer to Figure 3.92 in the text book.

Consider the expression for the voltage v_o.

$$v_o = i_1(10)$$

Therefore, the expression for the voltage v_o is, $i_1(10)$.

Step 2 of 4

Applying Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero.

For mesh 1, Kirchhoff's voltage law gives,

$$-40 + i_1(10) + (i_1 - i_2)(10) = 0$$
$$-40 + 10i_1 + 10i_1 - 10i_2 = 0$$
$$20i_1 - 10i_2 - 40 = 0$$
$$2i_1 - i_2 - 4 = 0$$

$i_2 = 2i_1 - 4$ …… (1)

Therefore, the expression for the current i_2 is, $2i_1 - 4$.

Step 3 of 4

For mesh 2, Kirchhoff's voltage law gives,

$$i_2(10)+(i_2-i_1)(10)+2v_o=0$$
$$10i_2+10i_2-10i_1+2v_o=0$$
$$20i_2-10i_1+2v_o=0$$

Substitute $i_1(10)$ for v_o and $2i_1-4$ for i_2.

$$20(2i_1-4)-10i_1+2i_1(10)=0$$
$$40i_1-80-10i_1+20i_1=0$$
$$50i_1-80=0$$
$$50i_1=80$$

$$i_1=\frac{80}{50}$$
$$=1.6\text{ A}$$

Therefore, the value of the mesh current i_1 is, $\boxed{1.6\text{ A}}$.

Comments (2)

Anonymous

why when I sub in 10i1 for V-not and solve for i1 in terms of i2 it gives me the wrong answer. I got i1 = -2i2. I would then sub that into the last equation to solve for i2.

Anonymous

never mind I got it

Step 4 of 4

Recall equation (1).

$$i_2=2i_1-4$$

Substitute 1.6 A for i_1.

$$i_2=2(1.6)-4$$
$$=3.2-4$$
$$=-0.8\text{ A}$$

Therefore, the value of the mesh current i_2 is, $\boxed{-0.8\text{ A}}$.